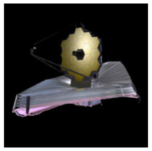

7827 - Looking into the core of the explosions of massive stars

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	24ggi	MIRI Low Resolution Spectroscopy	(1) 2024ggi
	3	24iss	MIRI Low Resolution Spectroscopy	(3) 2024iss
	4	24ahv	MIRI Low Resolution Spectroscopy	(4) 2024ahv
	8	23ixf	MIRI Low Resolution Spectroscopy	(8) 2023ixf
	9	23rve	MIRI Low Resolution Spectroscopy	(9) 2023rve

ABSTRACT

The physics governing the core-collapse supernova phenomenon is only indirectly constrained. Yet, knowledge of the yields of heavy elements synthesised in supernovae is a crucial ingredient in many areas of astrophysics. However, such measurements are challenging: the innermost regions -- where the heavy elements reside -- are only visible several hundred days after the explosion, when the ejecta have become optically thin, but the supernova has become faint. Thus, ground-based observations are generally unfeasible. The excellent sensitivity of JWST/MIRI finally allows us a chance to carry out such measurements, but only for the nearest of supernovae. We propose to seize a once-in-a-lifetime opportunity afforded by the fortuitous occurrence of nearby supernovae to take snapshots of the innermost regions of the explosions arising from massive stars and allowing us to directly measure yields of important species from lines that are only available in the mid-IR region. This is a crucial first step towards exploring the parameter space spanned by core-collapse supernovae and the stars or stellar systems that give rise to them. Remarkably, such datasets do not exist

for the variety of massive star supernova subtypes, and there is no prospect of deferring this exercise to future years. With our progam of MIRI observations we will take the first steps in this direction. With targets that have excellent ancillary data, including information on the progenitor star, a modest investment of JWST time will lead to the creation of datasets with true legacy value.

OBSERVING DESCRIPTION

We request MIRI-LRS observations of a small sample of core-collapse supernovae. The idea is to provide a

The above strategy ensures that we can access key spectral diagnostics, monitor their evolution, allowing us to robustly determine elemental masses and compare with predictions of explosion models of massive stars.

Proposal 7827 - Targets - Looking into the core of the explosions of massive stars

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	2024ggi	RA: 11 18 22.0870 (169.5920292d) Dec: -32 50 15.27 (-32.83758d) Equinox: J2000		
Comments: *Category=Star* *Description=[Supernovae]*				
(3)	2024iss	RA: 12 59 6.1330 (194.7755542d) Dec: +28 48 42.62 (28.81184d) Equinox: J2000		
Comments: *Category=Star* *Description=[Supernovae]*				
(4)	2024ahv	RA: 16 18 46.3100 (244.6929583d) Dec: +07 24 44.84 (7.41246d) Equinox: J2000		
Comments: *Category=Star* *Description=[Supernovae]*				
(8)	2023ixf	RA: 14 03 38.5620 (210.9106750d) Dec: +54 18 41.94 (54.31165d) Equinox: J2000		
Comments: *Category=Star* *Description=[Supernovae]*				
(9)	2023rve	RA: 02 46 18.1300 (41.5755417d) Dec: -30 14 22.16 (-30.23949d) Equinox: J2000		
Comments: *Category=Star* *Description=[Supernovae]*				

Observation

Proposal 7827, Observation 1: 24ggi Tue Mar 11 21:07:37 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Comments: There is a very short window (3 days) at the start of cycle 4. We have set "between dates" constraints for the longer window that opens in Dec. 2025 (and indeed used this epoch for the time estimates.

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	2024ggi	RA: 11 18 22.0870 (169.5920292d) Dec: -32 50 15.27 (-32.83758d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F560W	FASTGRPAVG	10	1	1	111.002	226436

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	20	2	2	1	1	113.777		F1000W

Proposal 7827 - Observation 1 - Looking into the core of the explosions of massive stars

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	100	20	40	1	2	11205.612	226436

Special Requirements

Between Dates 20-DEC-2025:00:00:00 and 16-JAN-2026:00:00:00

Observation

Proposal 7827, Observation 3: 24iss

Tue Mar 11 21:07:37 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Comments: Time variable target. Window is set to provide flexibility in scheduling while avoiding further fading of the target.

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	2024iss	RA: 12 59 6.1330 (194.7755542d) Dec: +28 48 42.62 (28.81184d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F560W	FASTGRPAVG	10	1	1	111.002	226436

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	20	2	2	1	1	113.777		F1000W

Proposal 7827 - Observation 3 - Looking into the core of the explosions of massive stars

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	100	15	30	1	2	8402.821	226436

Special Requirements

Between Dates 16-DEC-2025:00:00:00 and 16-JAN-2026:00:00:00

Observation

Proposal 7827, Observation 4: 24ahv

Tue Mar 11 21:07:37 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Comments: Time variable target. Window is set to provide flexibility in scheduling while avoiding further fading of the target.

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	2024ahv	RA: 16 18 46.3100 (244.6929583d) Dec: +07 24 44.84 (7.41246d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F560W	FASTGRPAVG	10	1	1	111.002	226436

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	20	2	2	1	1	113.777		F1000W

Proposal 7827 - Observation 4 - Looking into the core of the explosions of massive stars

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	100	15	30	1	2	8402.821	226436

Special Requirements

Between Dates 01-JUL-2025:00:00:00 and 01-AUG-2025:00:00:00

Observation

Proposal 7827, Observation 8: 23ixf

Tue Mar 11 21:07:37 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Comments: Time variable target. Window is set to provide flexibility in scheduling while avoiding further fading of the target.

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 8:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	2023ixf	RA: 14 03 38.5620 (210.9106750d) Dec: +54 18 41.94 (54.31165d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F560W	FASTGRPAVG	10	1	1	111.002	226436

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	20	2	2	1	1	113.777		F1000W

Proposal 7827 - Observation 8 - Looking into the core of the explosions of massive stars

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	100	15	30	1	2	8402.821	226436

Special Requirements

Between Dates 06-DEC-2025:00:00:00 and 07-JAN-2026:00:00:00

Observation

Proposal 7827, Observation 9: 23rve

Tue Mar 11 21:07:37 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Comments: Time variable target. Window is set to provide flexibility in scheduling while avoiding further fading of the target.

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 9:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(9)	2023rve	RA: 02 46 18.1300 (41.5755417d) Dec: -30 14 22.16 (-30.23949d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F560W	FASTGRPAVG	10	1	1	111.002	226436

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
1	FASTR1	20	2	2	1	1	113.777		F1000W

Proposal 7827 - Observation 9 - Looking into the core of the explosions of massive stars

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	FASTR1	100	15	30	1	2	8402.821	226436

Special Requirements

Between Dates 11-JUL-2025:00:00:00 and 05-OCT-2025:00:00:00